

Bike Sharing Demand Prediction Using Multiple Linear Regression

This project builds a multiple linear regression model to predict daily demand for shared bikes for BoomBikes, a bike-sharing provider seeking to understand the factors that drive ridership and shape its post-pandemic recovery strategy. The analysis draws on 730 daily observations spanning 2018–2019, covering 16 variables including weather conditions, temperature, humidity, windspeed, season, and calendar features. Daily rentals averaged 4,508 rides, ranging from a low of 22 to a peak of 8,714.

The dataset required no imputation, containing no missing values or duplicates. Categorical variables such as season, month, weekday, and weather situation were converted into dummy variables using `drop_first=True` to prevent multicollinearity, expanding the feature space to 32 model-ready columns. The data was split 70/30 into 511 training and 219 test observations, with `MinMaxScaler` applied to numeric features. Exploratory analysis revealed that temperature was the numeric feature most strongly correlated with demand ($r \approx 0.63$), while average daily rentals surged from 3,406 in 2018 to 5,610 in 2019 — a 65% year-over-year increase.

Feature selection combined Recursive Feature Elimination, which shortlisted 15 candidate variables, with manual backward elimination guided by p-values and Variance Inflation Factors in `statsmodels OLS`. The final model retained nine statistically significant features and satisfied all core regression assumptions: residuals were approximately normally distributed and homoscedastic, and a Durbin–Watson statistic of 2.16 confirmed independence of errors.

The model explains demand well, achieving an R^2 of 0.816 (adjusted R^2 0.812) on training data and 0.813 on unseen test data — a gap of just 0.3 percentage points, demonstrating excellent generalisation with no overfitting. The F-statistic of 246.5 ($p < 0.001$) confirms overall model significance. Temperature emerged as the strongest positive driver (coefficient +3,984), followed by year (+1,995), while bad weather (−2,256) and the spring season (−1,168) suppressed demand most sharply; September (+466) stood out as a peak month.

For BoomBikes, the strategic implications are clear: concentrate fleet capacity and marketing in warm months, anticipate sharp demand drops on poor-weather days with dynamic pricing, and invest in capacity expansion to capture the strong underlying growth trend in ridership.